

Wilka Torrico De Carvalho, *Reinforcement Learning Researcher and Cognitive Scientist*

CONTACT INFORMATION	Website: cogscikid.com	Github: github.com/wcarvalho
	E-mail: wcarvalho@g.harvard.edu	Google Scholar
EMPLOYMENT	Harvard , Boston, MA Kempner Fellow	Sept 2023 –
	DeepMind , London, UK Research Scientist Intern	Sept 2022 – May 2023
	DeepMind , London, UK Research Scientist Intern	Mar 2021 – July 2021
	Microsoft Research , Redmond, WA Research Scientist Intern	June 2018 – Aug 2018
	IBM Research , San Jose, CA Research Scientist Intern	Sept 2017 – Dec 2017
EDUCATION	VISA Research , Palo Alto, CA Research Scientist Intern	May 2017 – Aug 2017
	University of Michigan–Ann Arbor , Ann Arbor, MI <i>School of Engineering</i> , Ph.D. in Computer Science Advisors: Satinder Singh , Honglak Lee , Richard Lewis	Sep 2018 - July 2023
	University of Southern California , Los Angeles, CA <i>Viterbi School of Engineering</i> , M.S. in Computer Science Advisor: Yan Liu	Aug 2015 - May 2017
	Stony Brook University , Stony Brook, NY <i>College of Arts and Sciences</i> , B.S. in Physics Advisor: Axel Drees	Aug 2011 - May 2015
	Brooklyn Technical High School , Brooklyn, NY Diploma in Applied Physics	May 2007 - May 2011
HONORS & AWARDS	GitLab Foundation AI for Economic Opportunity Award Finalist (7% selected)	2025
	Kempner Institute Fundamentals of Intelligence Fellowship	2023
	ICML, NeurIPS Top Reviewer	2022, 2023
	University of Michigan “The Glass is Half-Full” Award	2019
	Hertz Fellowship Finalist (10% selected)	2019
	1/200 chosen internationally for Heidelberg Laureate Forum	2018
	GEM National Fellowship sponsored by IBM, Adobe	2017, 2018
	University of Michigan Rackham Merit Fellowship (12% selected)	2017
	ICLR, Neurips BAI Travel Award	2017, 2019
	NSF Graduate Research Fellowship (Neuroscience)	2015
	Provost Award for Academic Excellence (20/5000 graduates chosen)	2015
	Researcher of the Month (1 monthly at Stony Brook University)	2014
	HHMI Minority Undergraduate Research Fellowship	2014
	ΣΠΣ Physics Honor Society (sponsored by Alfred Goldhaber)	2013

PREPRINTS

Wilka Carvalho, Sam Hall-McMaster, Honglak Lee, Samuel J. Gershman. “*Preemptive Solving of Future Problems: Multitask Preplay in Humans and Machines.*” 2025

Wilka Carvalho*, Andrew Lampinen*. “*Naturalistic Computational Cognitive Science.*” 2025

Aneesh Muppidi, Samuel J. Gershman, **Wilka Carvalho**. “*Variational Agent Discovery.*” 2025

PUBLICATIONS

Kunal Jha, **Wilka Carvalho**, Natasha Jaques, Max Kleiman-Weiner. “*Cross-environment Cooperation Enables Zero-shot Multi-agent Coordination.*” In **ICML**, 2025 (**Oral**)

Wilka Carvalho*, Momchil S Tomov*, William de Cothi*, Caswell Barry, Samuel J. Gershman. “*Predictive representations: building blocks of intelligence.*” In **Neural Computation**, 2024

Wilka Carvalho, Andre Saraiva, Angelos Filos, Andrew Lampinen, Loic Matthey, Richard L Lewis, Honglak Lee, Satinder Singh, Danilo Jimenez Rezende, Daniel Zoran. “*Combining behaviors with the successor features keyboard.*” In **NeurIPS**, 2024

Wilka Carvalho, Andrew Lampinen, Kyriacos Nikiiforou, Felix Hill, Murray Shanahan. “*Feature-Attending Recurrent Modules for Generalization in Reinforcement Learning.*” In **TMLR**, 2023

Wilka Carvalho, Angelos Filos, Richard L. Lewis, Honglak Lee, Satinder Singh. “*Composing Task Knowledge with Modular Successor Feature Approximators.*” In **ICLR**, 2023

Lajanugen Logeswaran, **Wilka Carvalho**, Honglak Lee. “*Learning Compositional Tasks from Language Instructions.*” In **AAAI**, 2023

Chris Hoang, Sungryull Sohn, Jongwook Choi, **Wilka Carvalho**, Honglak Lee. “*Successor Feature Landmarks for Long-Horizon Goal-Conditioned Reinforcement Learning.*” In **NeurIPS**, 2021

Wilka Carvalho, Anthony Liang, Kimin Lee, Sungryull Sohn, Honglak Lee, Richard L. Lewis, Satinder Singh. “*Reinforcement Learning for Sparse-Reward Object-Interaction Tasks in First-person Simulated 3D Environments.*” In **IJCAI**, 2021

Wilka Carvalho*, Sanjay Purushotham*, Tanachat Nilanon, Yan Liu. “*Variational Recurrent Adversarial Domain Adaptation.*” In **ICLR**, 2017

WORKSHOP
PUBLICATIONS

Wilka Carvalho, Vikram Goddla, Ishaan Sinha, Hoon Shin, Kunal Jha. “*NiceWebRL: a Python library for human subject experiments with reinforcement learning environments.*” In Human-AI Complementarity for Decision Making Workshop, 2025 (**Oral**)

Wilka Carvalho, Vikram Goddla, Ishaan Sinha, Hoon Shin, Kunal Jha. “*NiceWebRL: a Python library for human subject experiments with reinforcement learning environments.*” In New York RL Workshop, 2025 (**Oral**)

Lajanugen Logeswaran, **Wilka Carvalho**, Honglak Lee. “*Learning Compositional Tasks from Language Instructions.*” In NeurIPS Deep RL Workshop, 2021

Wilka Carvalho, Murray Shanahan. “*Learning to Represent State with Perceptual Schemata.*” In ICML {Unsupervised RL, Real Life RL} Workshops, 2021

Wilka Carvalho, Anthony Liang, Kimin Lee, Sungryull Sohn, Honglak Lee, Richard L. Lewis,

Satinder Singh. “*Reinforcement Learning for Sparse-Reward Object-Interaction Tasks in First-person Simulated 3D Environments.*” In {ICML Object-Oriented Learning, NeurIPS DeepRL} Workshops, 2020

Chris Hoang, Sungryull Sohn, Jongwook Choi, **Wilka Carvalho**, Honglak Lee. “*Successor Landmarks for Efficient Exploration and Long-Horizon Navigation.*” In NeurIPS DeepRL Workshop, 2020

Wilka Carvalho, Kimin Lee, Anthony Liang, Ryan Krueger, Richard L. Lewis, Satinder Singh, Honglak Lee. “*Efficiently Learning to Perform Household Task with Object-oriented Exploration.*” In NeurIPS BAI, 2019 (**Oral**)

Bryant Chen*, **Wilka Carvalho***, Benjamin Edwards, Taesung Lee, Ian Molloy, Heiko Ludwig. “*Detecting Backdoor Attacks on Deep Neural Networks by Activation Clustering.*” In AAAI Artificial Intelligence Safety Workshop, 2018 (**Best Paper**)

Sanjay Purushotham*, **Wilka Carvalho***, Yan Liu. “*Variational Adversarial Deep Domain Adaptation for Health Care Time Series Analysis.*” In NeurIPS Machine Learning for Healthcare Workshop, 2016 (**Spotlight**)

Wilka Carvalho. “*Modeling a Detection of internally reflected Cherenkov light (DIRC) Particle Detector for High-Multiplicity Collisions.*” In SURC, 2015

PATENTS

Wilka Carvalho, Angelos Filos, Danilo Jimenez Rezende, Daniel Zoran. “*Controlling Agents by Transferring Successor Features To New tasks.*” 2023

Wilka Carvalho. “*Compositional Generalization in Reinforcement Learning.*” 2023

Bryant Chen, **Wilka Carvalho**, Heiko Ludig, Ian Molloy, Jialong Zhang, Benjamin Edwards. “*Detecting poisoning attacks on neural networks by activation clustering.*” 2020

Wilka Carvalho, Bryant Chen, Benjamin Edwards, Taesung Lee, Ian Molloy, Jialong Zhang. “*Using Gradients to Detect Backdoors in Neural Networks.*” 2018

Wilka Carvalho, Yan Liu, Tanachat Nilanon, Sanjay Purushotham. “*Effective Knowledge Transfer Among Patient Populations via Deep Learning.*” 2017

INVITED TALKS

NiceWebRL: A Python Library for Human Experiments with Reinforcement Learning Environments

Human-AI Complementarity for Decision Making Workshop. *Carnegie Mellon University*. (September, 2025)

New York Reinforcement Learning Workshop. *NYC*. (September, 2025)

Cognitive Computational Neuroscience Conference. *Amsterdam*. (August, 2025)

Naturalistic Computational Cognitive Science

Cognitive Computational Neuroscience Conference. *Amsterdam*. (August, 2025)

Scaling Up Reinforcement Learning Theories of Cognition

Cognitive Science Colloquium Series. *University of Maryland at College Park*. (December, 2026)

Machine Intelligence through Decision-making and Interaction Lab. *UT Austin*. (September, 2025)

Cognition, Brain, and Behavior Seminar. *Harvard University*. (April, 2025)

Center for Theoretical Neuroscience. *Columbia University*. (April, 2025)

Mattar Lab. *NYU*. (April, 2025)

Emerge Lab. *NYU*. (April, 2025)
 Computation and Decision-Making Lab. *NYU*. (April, 2025)
 RL Seminar. *Tübingen*. (March, 2025)
 Google DeepMind. *San Francisco*. (March, 2025)
 Cognition Seminar. *UC Berkeley*. (March, 2025)
 The Rational Altruism Lab. *UCLA*. (March, 2025)
 Social and Decision Neuroscience Seminar. *Caltech*. (March, 2025)
 Symposium on “Frontiers of Machine Learning and AI. *USC*. (March, 2025)
 Neurolunch Seminar. *Harvard*. (March, 2025)
 RL Seminar. *UMich*. (February, 2025)
 Vision Lab. *Harvard University*. (February, 2025)
 Sabatini Lab. *Harvard University*. (February, 2025)

Predictive Representations: Building Blocks of Intelligence

Computational Cognitive Science Laboratory. *UC Berkeley*. (March, 2025)
 Computational Cognitive Neuroscience Lab. *Harvard University*. (February, 2025)
 Neuro Systems Club. *Harvard University*. (Fall, 2024)
 Mathis Group for Computational Neuroscience and AI. *EPFL*. (Fall, 2024)
 Theory of Learning Lab. *Gatsby Computational Neuroscience Unit (UCL)*. (Summer, 2024)
 Conference on the Mathematics of Neuroscience and AI. *Rome, Italy*. (Summer, 2024)
 Theoretical Cognitive Neuroscience Lab. *Boston University*. (Fall, 2023)

Towards Human-like Deep Reinforcement Learning

Computational Cognitive Science Group. *Princeton*. (Fall, 2023)
 Kempner Institute. *Harvard University*. (March, 2023)

Composing Task Knowledge with Modular Successor Feature Approximators

RL at Harvard. *Harvard University*. (August, 2023)

Weak Inductive Biases for Composable Primitive Representations

Computational Neuroscience Group. *UCL*. (October, 2022)
 Intelligent Robot Lab. *Brown*. (February, 2022)
 Computational Cognitive Neuroscience Lab. *Harvard University*. (February, 2022)
 Computational Cognitive Neuroscience Lab. *University of California—Berkeley*. (January, 2022)
 Deep Learning Community of Practice. *Intel*. (January, 2022)
 Deep RL Workshop. *NeurIPS*. (December, 2021)

Reinforcement Learning for Sparse-Reward Object-Interaction Tasks in a First-Person Simulated 3D Environment

Object Representations for Learning Workshop. *NeurIPS*. (December, 2020)
 Cognitive Tools Lab. *University of California—San Diego*. (November, 2020)
 Cognitive Science Community. *University of Michigan*. (November, 2020)
 Stony Brook Society of Physics. *Stony Brook, NY*. (April, 2020)

Variational Recurrent Adversarial Deep Domain Adaptation

Machine Learning Lunch Seminar. *University of Southern California*. (April, 2017)

TEACHING EXPERIENCE

Harvard University, Cambridge, MA
Teaching Assistant, *Fall 2025*
 Building Transformers from Scratch Workshop

Stony Brook University, Stony Brook, NY
Calculus Instructor, *Spring 2015*

Worked with two math professors to develop and teach a supplementary calculus curriculum that promoted minority representation in stem majors.

Stony Brook University, Stony Brook, NY

Educational Opportunity Program Personal Tutor, *Spring 2013 - Fall 2014*

Tutored marginalized students in introductory physics and math courses

SERVICE

Action Editor, Transactions on Machine Learning Research, 2023 -

Area Chair, Reinforcement Learning Conference, 2023 -

Reviewer, Royal Society Open Science, 2025 -

Reviewer, Journal on Machine Learning Research, 2025 -

Reviewer, Perspectives on Psychological Science, 2025 -

Reviewer, Neural Computation, 2023 -

Reviewer, ICLR, 2021 -

Reviewer, ICML, 2021 -

Reviewer, NeurIPS, 2020 -

Student Volunteer, ICLR, 2017

OUTREACH

Lecture for summer program for disadvantaged students, Dorchester, MA, 2024

DAPCEP (Detroit area pre college engineering program), Michigan, 2023

Stony Brook's CSTEP Alumni mentoring program, Michigan, 2022

HS2 Engineer's Highschool Panel, Virtual, 2021

Stony Brook Society of Physics, Stony Brook, NY, 2020

Michigan Explore Graduate Studies Workshop, Ann Arbor, MI, 2019

Research and Fellowships Week NSF Panel, Los Angeles, CA, 2016

National Society of Black Engineers Grad Panel, Los Angeles, CA, 2016

Graduate School External Fellowship Boot Camp, Los Angeles, CA, 2016

Mentored marginalized high school youth through the Pullias Center for Higher Education, Los Angeles, CA, 2016

Engineering Graduate Diversity Symposium, Los Angeles, CA, 2015

Black Student Association: What it takes to go to Graduate School, Los Angeles, CA, 2015

Collegiate Science and Technology Entry Program Undergraduate Research Panel, Stony Brook, CA, 2014

SKILLS

Machine Learning Software: Pytorch, TensorFlow, Theano, Keras

Neuroscience Software: Neuron

Languages: Python, C++, C, Java

Systems: Unix, Linux, OSX

PRESS

[Finding Gaps in Your Grad School Apps](#)

[This Week In Machine Learning \(TWIML\) Research Feature](#)

[University of Michigan Research Feature](#)

[Exploring the source of social stereotypes](#)

[Black History Month: Why a career in science?](#)

[Research Feature by the USC Graduate School](#)

[2015 NSF Graduate Research Fellow Wilka Carvalho](#)

[Biomath Learning Center Launches Modified Supplemental Instruction Program](#)

[URECA Research of the Month: Wilka Carvalho](#)

[Student Feature by Stony Brook University](#)

INTERESTS

• traveling • chess • software development • improvisational dance • deadpan humor